



BABA FARAH

INGÉNIEURE EN SYSTÈMES EMBARQUÉS & ROBOTIQUE

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Recent Master's graduate in Computer Engineering and Embedded Systems, with hands-on experience in robotics, IoT, embedded systems, sensors, and data acquisition. Experienced with ESP32, Raspberry Pi, C/C++, Python, and embedded system prototyping. Strong interest in robotics, edge AI, computer vision, and connected agricultural systems.

EDUCATION

- Master of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2024–2026)
- Bachelor of Excellence in Computer Engineering and Embedded Systems, Ibn Zohr University – FSA, Agadir (2023–2024)
- Bachelor in Education – Computer Science, Higher School of Education and Training, Agadir (2020–2023)
- Baccalaureate in Physical Sciences (French option), Lycée Libres Agadir Ida Outanane (2022)
- Baccalaureate in Physical Sciences (Arabic option), Lycée Mohammed Derfoufi, Agadir (2019)

PROFESSIONAL EXPERIENCES

Embedded Systems Engineer – Final-Year Internship | OCP Jorf Lasfar | Feb. – Jun. 2026

- Designed and developed an embedded system for automated acquisition and processing of physical data.
- Integrated sensors, embedded devices, and electronic components within the system architecture.
- Analyzed technical data and defined processing requirements for the embedded solution.
- Contributed to system modeling, testing, and validation.
- Worked on the integration of hardware and software components to ensure system consistency and reliability.
- Documented technical aspects of the developed solution.

Software Developer – Internship | MARSA MAROC, Agadir | Aug. – Sep. 2025

Developed a web application for SQL query management and execution using Laravel and MySQL.

Software Developer – Internship | National Ports Agency (ANP), Agadir | Jul. 2024

Developed a web application for IT asset management using Laravel and MySQL.

RECENT TECHNICAL PROJECTS

Embedded System & Programmable Logic (FPGA / PCB) | Nov. 2025

Architectural analysis of electronic circuits and synthesis on programmable logic boards (CPLD/FPGA – VHDL), combined with hardware integration, PCB design and routing, soldering, and validation using operational test benches.

Tools: VHDL, C/C++, Microcontrollers (STM32/PIC/Arduino), PCB Design

HealthTech Hackathon – Context-Aware Cardio-Motor Anomaly Monitoring | Dec. 2025

Designed a smart wearable bracelet for the acquisition and monitoring of cardio-motor data.

Tools: ESP32, MAX30102, MPU6050, IoT, Machine Learning, Flutter, Firebase

Autonomous Mobile Surveillance Robot | Nov. 2025

- Designed an embedded robotic system for autonomous movement and environmental monitoring.
- Integrated microcontroller, sensors, camera, motors, and electronic components.
- Worked on the interaction between hardware and software components.
- Technologies: ESP32, ESP32-CAM, motion sensors, camera, motors.

Tools: Esp32, Esp32-CAM, motion sensors, camera, motors

TECHNICAL SKILLS

Robotics & Embedded Systems: ESP32, Raspberry Pi, Microcontrollers, Embedded Systems, Robotics, System Integration, Prototyping

Programming: C, C++, Python, Assembly, VHDL, Python, Java

Sensors & Actuators: Motion Sensors, Temperature Sensors, Environmental Sensors, Cameras, Motors, Actuators, Data Acquisition

IoT & Communication: IoT, MQTT, Client-Server Architecture, Connected Devices

Computer Vision & AI: Computer Vision, Image Processing, Machine Learning

Tools & Platforms: MATLAB, Proteus, Quartus, Git/GitHub, Postman, Linux

Web & Databases: SQL, MySQL, Firebase, Flask, React

CERTIFICATIONS

- Technovate Hackathon AgriTech / HealthTech
- Automotive Engineering: Testing Strategies, UDS & Functional Safety (ISO 26262)

LANGUAGES

Arabic : Native ; **French :** Fluent ; **English :** Fluent ; **Russian :** Intermediate